

The Demand Side of the Trust Stack: what governments procure to secure, identify and decide, measured against supply, risk and the post-quantum mandate

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Draft v2, 2026-07-04. Data: Zenodo 10.5281/zenodo.21192405 (CC-BY-4.0). Code and linked observatories: <https://github.com/tyche-dev/trust-tech-procurement-observatory>. Companion article: the Quantum-Readiness Procurement Gap (Sokolov 2026b).

Abstract

Digital government rests on a stack of trust technologies: public-key infrastructure and the post-quantum cryptography that must succeed it, the digital identity that establishes who may act, and the artificial intelligence that increasingly makes the decisions in between. We read that stack from its demand side, through what governments actually procure, and we measure procurement against three other observable quantities: the supply of qualified trust services, the record of AI incidents, and the mandate to migrate to quantum-safe cryptography. Using a live, provenance-verified corpus of 4,401 public-sector procurement records across the four domains and 106 countries, and linking it (without fusing it) to a census of qualified remote-signature trust services and to a catalogue of 998 reviewed AI incidents, we report four results organised by one finding: the state buys capability far ahead of assurance across the stack. The assurance gap is directly measurable in the artificial-intelligence layer, where only 3.72 percent of AI procurement records carry any assurance term (audit, conformity, risk or impact assessment, safety, evaluation, certification, or governance among them). Its sharpest instance is cryptographic: post-quantum cryptography, the assured replacement for today’s public-key trust, is procured at one fifty-seventh of the quantum-vulnerable base (2,846 public-key-infrastructure and digital-identity records against 50 post-quantum) — a cliff of 56.9 to one that widens to 118.6 on a stricter confirmed count and to 90.4 by United States federal award value. This is the companion article’s quantum-readiness gap, here situated inside the wider stack and set against a datable installed base of qualified trust services that all require migration; because it is the cryptographic face of the same assurance ratio, it restates rather than independently confirms the artificial-intelligence measure. Two further linkages qualify the picture. Trust-service supply is highly concentrated: on a services-count basis, within one census slice of the eIDAS trusted list, a single member state holds 68 percent of the remote qualified-signature market. And demand is decoupled from risk: across the 21 jurisdictions observable in both the procurement and incident records we find no detectable association between AI procurement volume and AI-incident count (Spearman $\rho = -0.13$, $n = 21$, $p \sim 0.58$), so procurement is not, at this resolution, steering toward where harm has already appeared. The procurement record makes each of these quantities observable before deployment, and every figure is recomputable from an open pipeline.

1. Introduction

A modern state runs on delegated trust. When a citizen files a tax return, when a benefit is granted or refused, when a document is signed on behalf of an agency, the transaction depends on a layered infrastructure: cryptography that authenticates a machine or a document, identity systems that decide who is authorised to act, and, increasingly, automated systems that make or shape the decision. These layers are procured. They appear, before they appear anywhere else, as public

procurement notices and contract awards that name the buyer, the object, and often the value.

We take the procurement record as an instrument for reading the state’s investment in its own trust stack, and we ask four questions that the record can answer when it is joined to what else is observable. The animating question is whether the state is buying the trustworthy version of each technology at a rate commensurate with the base technology it is deploying. The sharpest instance is the migration to post-quantum cryptography, which policy has scheduled with deadlines (National Institute of Standards and Technology 2024b, 2024a, 2024c; The White House 2022; Office of Management and Budget 2022; European Commission 2024) and which the companion article studies in isolation (Sokolov 2026b). Here we place that instance inside the wider stack and against three external references: the supply of qualified trust services, the record of where artificial intelligence goes wrong, and the demand for the assured version of each layer.

The four research questions are concrete. **RQ-1 (the quantum-trust cliff)**: what is the ratio of quantum-safe to quantum-vulnerable trust procurement, and how large is the installed base that must migrate? **RQ-2 (trust-supply concentration)**: how concentrated is the supply of qualified trust services relative to the demand? **RQ-3 (demand-risk alignment)**: do governments procure artificial intelligence in the jurisdictions where AI incidents concentrate? **RQ-4 (the assurance gap)**: across the layers of the stack, what share of procurement buys the assured version of the technology rather than the base? We answer each with real, recomputable numbers, and we are explicit about where a coverage limit makes a result preliminary rather than definitive.

The methodological commitment throughout is to **link but never fuse** (Sokolov 2026b). Each result joins the procurement observatory to one other dataset by an explicit key, preserves the provenance of both, and copies no field from one observatory into the other. This keeps every claim traceable to a source that the reader can check, which matters especially for a study whose central finding is an absence.

2. Background and related work

The trust stack and its observatories. The four domains are the procurement-visible layers of one stack. Public-key infrastructure is the incumbent cryptographic root; post-quantum cryptography is the migration that keeps that root from collapsing under a quantum attack; digital identity, governed in the European Union by the eIDAS framework and its 2024 revision (European Union 2024a), is the layer that establishes who may act; and artificial intelligence, now the subject of horizontal regulation (European Union 2024b), is the decision layer above. We observe each from a different angle: procurement for demand, a qualified-trust-service census for identity supply, and a reviewed catalogue of AI incidents (Sokolov 2026a) for failure.

Prior measurement of the post-quantum transition has been almost entirely deployment-side. Network telemetry counts the share of encrypted connections that negotiate a post-quantum key exchange, a share that has moved from negligible to a rising minority of TLS traffic as browsers and content networks turned on hybrid key agreement (Westerbaan et al. 2024, 2025), and readiness surveys probe how far server and library support has spread (Dubey, Varshney, et al. 2026), while the threat model that gives the deadline its urgency, harvest-now-decrypt-later, is now under formal feasibility study (Blanco-Romero et al. 2026). Deployment telemetry answers a real question, whether the wire is protected today, but it is blind to what the state has committed to buy and therefore to what will or will not be protected in the migration window that policy has scheduled. Procurement precedes deployment by months or years, and it is the earliest public signal of that commitment; it is the signal this article reads, and it is unread in the post-quantum literature.

Procurement microdata as a policy signal is established in the innovation and public-management literature, where award records have been used to study demand-side innovation policy (Demircioglu and Vivona 2021) and, more recently, to characterise government demand for artificial intelligence specifically (Hickok 2024; Johnson et al. 2024). Open award databases now make the raw material tractable at national scale (Potin et al. 2023; Omari et al. 2025), and the Open Contracting Data Standard (Open Contracting Partnership 2021) gives the records a shared model. What that work has not done is read the trust stack as a stack, across the cryptographic, identity and artificial-intelligence layers at once, and join the demand signal to what is observable on the other sides of each layer.

On the other sides sit two very different kinds of record. Qualified trust services in the European Union are enumerated in national trusted lists under the eIDAS framework, a supply register whose market structure is known to be uneven and whose unit of analysis is subtle, since listed certificate entries inflate a count of distinct services; a careful reading reports the funnel from entries to services. The record of AI harm, by contrast, is an incident catalogue assembled from public reporting, and it inherits that provenance: incidents are recorded where journalists, regulators and researchers are looking, so the catalogue is a map of observed harm rather than of harm, and any jurisdictional comparison built on it carries a reporting bias that must be stated. Reading procurement against a concentrated supply register on one side and a reporting-biased incident catalogue on the other is precisely why the linkage discipline below refuses to fuse the datasets: each retains the caveats of its own construction. That cross-observatory demand view, with its caveats kept attached, is the gap this article fills.

3. Data and methods

The procurement observatory (demand). A live, provenance-verified corpus of 4,401 unique records across artificial intelligence, post-quantum cryptography, public-key infrastructure and eIDAS or digital identity, harvested from four public sources (EU Tenders Electronic Daily, US USAspending, UK Contracts Finder and Find a Tender, and World Bank procurement notices) and spanning 106 countries. A record exists only if a deterministic harvest script received a live successful response from a listed source, and every record carries its source URL, fetch timestamp, response status and a hash of the raw payload; a validator rejects any record lacking provenance, carrying a future date, or naming a placeholder host. The companion article documents the corpus and the anti-fabrication construction in full (Sokolov 2026b). By domain the corpus holds 1,727 public-key-infrastructure, 1,505 artificial-intelligence, 1,119 eIDAS and 50 post-quantum records.

The linked observatories. For supply we use a census of 114 qualified remote signature and seal creation-device management services across ten European member states (snapshot 2026-07-04). For failure we use a catalogue of 998 reviewed AI incident and vulnerability records spanning 116 jurisdictions (snapshot 2026-07-04) (Sokolov 2026a), each carrying a jurisdiction, an AI-system type and a severity. Both are snapshotted read-only with a provenance manifest, each carrying a SHA-256 of its payload, and are linked to, never merged into, the procurement corpus.

Linkage. Every result in this article rests on exactly two deterministic operations and no others. The first is an exact-tier country join: each record on both sides is normalised to an ISO-3166 alpha-2 code (from its three-letter code or its country name) and joined on that key, which is what links procurement to the trust-service census (RQ-1, RQ-2) and to the incident catalogue (RQ-3). The second is in-corpus keyword or text matching: domain tags are assigned within the procurement corpus, and the assurance measure of RQ-4 counts assurance terms in the record

text. There is no record-to-record entity resolution here; we copy no field from one observatory into another. The heavier machinery of confidence-tiered record linkage, exact-identifier, strong-name, keyword-plausible and unmatched with a positive-control set, is what a fuller study needs for the full trusted-list supply (a complete RQ-2) and for the procurement-to-register join that our companion registered report addresses; those steps are pre-registered before data collection and are **not** used for any number reported below. Every reported figure is computed by an open analysis script from the committed data.

4. Results

4.1 RQ-1: the quantum-trust cliff

Governments procure the quantum-vulnerable trust layer, public-key infrastructure and digital identity together, at 2,846 records, against 50 records for its quantum-safe replacement: a ratio of **56.9 to one**. On the stricter confirmed post-quantum count of 24 (the companion article audits the tag and finds a post-quantum term in the retrievable text for 24 of the 50), the ratio is **118.6 to one**. In United States federal award value, where money is measured, public-key infrastructure outweighs post-quantum cryptography by **90.4 to one** (USD 566.7 million against 6.3 million). The vulnerable base is not hypothetical: the qualified-trust-service census enumerates **114 remote qualified signature and seal services across ten member states**, each running on today’s public-key cryptography and falling under the migration horizon set by the standards and mandates cited above. The state is buying the technology that must be replaced at roughly sixty to a hundred times the rate it buys the replacement, while operating a datable base that all requires migration. These record counts and dollar figures are the same underlying numbers the companion article reports for the post-quantum gap in isolation (Sokolov 2026b); they are not re-harvested here, and RQ-1’s contribution is to place them against the supply census and inside the wider stack rather than to re-measure them.

4.2 RQ-2: trust-supply concentration (preliminary)

Within the market for remote qualified signature and seal services, supply is extraordinarily concentrated: **a single member state holds 78 of 114 services, 68 percent**, with the remainder spread thinly across nine others. Digital-identity procurement demand, by contrast, is distributed across many member states. We report this as preliminary rather than definitive because the census covers one slice of the eIDAS trusted list, remote qualified signature and seal services only, so a naive reading that other states have no qualified trust supply would be an artifact of census coverage, not a finding. The defensible observation is the concentration within the measured slice; extending it to the demand-versus-supply map for the full trusted list is the natural next step and is method-ready.

4.3 RQ-3: demand and risk are decoupled

Across the 21 jurisdictions observable in both the procurement corpus and the incident catalogue, the rank correlation between artificial-intelligence procurement volume and AI-incident count is **Spearman rho = -0.13**. At this sample size that coefficient is statistically indistinguishable from zero ($t = -0.57$ on 19 degrees of freedom, $p \sim 0.58$), so the honest statement is not that demand runs against risk but that we find **no detectable association** between the two: procurement is not, at the country level and on present coverage, a leading indicator of incident exposure. The heavy procurers in the joined set are European, a reflection of the depth of the European tender

feed, whereas incidents concentrate in the United States, Russia and China. We flag that the two observatories have divergent coverage, procurement weighted toward the European Union and incidents toward the United States and globally, and that AI-incident catalogues are themselves subject to reporting bias (harm is recorded where journalists, regulators and researchers look, not uniformly across jurisdictions), so both the absence of a correlation and any apparent sign of it must be read against that confound. The result is a null at country resolution, to be tested properly by a within-jurisdiction, category-level crosswalk on a larger matched set.

4.4 RQ-4: the assurance gap

The state buys the assured or secured version of a technology at a small share of the base. The independent measure is in the artificial-intelligence layer, where only **3.72 percent** of AI procurement records carry any assurance term (audit, assurance, conformity, risk or impact assessment, safety, red-team, evaluation, certification, monitoring, bias, explainability, trustworthiness or governance). The cryptographic layer gives the same picture more starkly: post-quantum cryptography is **1.76 percent** of the trust base it must protect (0.84 percent on the confirmed count). We are explicit that this cryptographic figure is not a second, independent reading — 1.76 percent is exactly the reciprocal of the 56.9-to-one cliff of RQ-1 ($1 / 56.9$), so it re-expresses that result as a share rather than confirming it anew. What RQ-4 adds is that the same under-buying of assurance is visible in a second, unrelated layer, artificial intelligence, measured by a different operation on a different corpus. Across both, the assured share sits below four percent, with the cryptographic layer the more extreme. This is the through-line of the results: the state buys capability well ahead of assurance across the trust stack.

Figure. The demand side of the trust stack: three cross-observatory results.

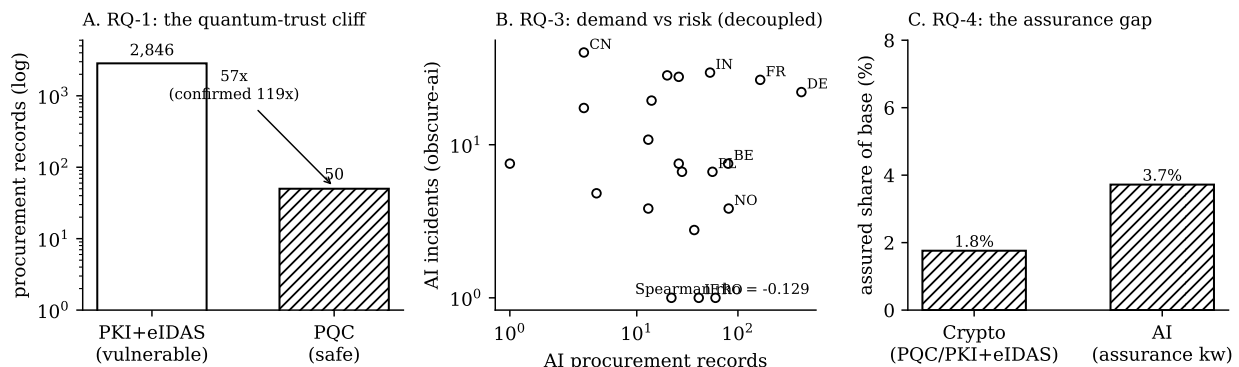


Figure 1: **Figure 1. The demand side of the trust stack, in three cross-observatory results.** (A) RQ-1, the quantum-trust cliff: quantum-vulnerable trust procurement (public-key infrastructure and eIDAS) against quantum-safe (post-quantum cryptography), log scale, with the ratio annotated. (B) RQ-3, demand against risk: AI procurement volume against AI-incident count per jurisdiction, showing no positive association (Spearman rho = -0.13). (C) RQ-4, the assurance gap: the assured share of the base technology in the cryptographic and artificial-intelligence layers. Every value is computed from the linked datasets; sources are linked, not fused.

5. Discussion

The four results describe one behaviour from four angles. Each is an association read from linked records, and we make no causal claim: the numbers describe what the state buys and against what, not why. The state is procuring the trust stack at scale, and it is procuring the assured version of each layer far more slowly than the base. The quantum-trust cliff (RQ-1) is the extreme case of the assurance gap (RQ-4): the assured cryptography is being out-procured by the very infrastructure it is meant to replace, against a fixed migration deadline and a threat model that front-loads that deadline. The decoupling of demand from risk (RQ-3) says that the under-buying of assurance is not being corrected by exposure: governments are not steering AI procurement toward the categories or places where harm has already appeared. The concentration of trust-service supply (RQ-2), even measured in one slice, warns that the assurance that is procured may rest on a narrow supplier base.

The policy implications separate cleanly by result and actor. For the quantum-trust cliff, the migration owners can convert a distant deadline into a near-term purchasing obligation by requiring agencies to report quantum-safe procurement and not only cryptographic inventory. These owners differ by jurisdiction: in the United States the budget office that issued the migration-reporting memorandum (Office of Management and Budget 2022), and in the European Union the coordinated-transition recommendation to member states (European Commission 2024). For the assurance gap in artificial intelligence, the corrective is a procurement one: an explicit assurance line item, tied to the obligations of the AI regulation (European Union 2024b), would make assured-versus-base demand auditable. For supply concentration, the trusted-list and single-market authorities have an interest in whether the qualified-service base is broad enough to carry the demand that eIDAS 2.0 (European Union 2024a) will generate. In every case the procurement record is the instrument that makes the quantity observable before deployment, complementary to the deployment telemetry on which readiness has until now been measured (Westerbaan et al. 2025).

6. Limitations

Coverage is a curated demand-signal sample, not a census of all government procurement; the European feed is the deepest and the United States and United Kingdom feeds undercount, so cross-region comparisons are of visible procurement, and our central ratios are within-region or within-source where possible. The link-not-fuse discipline is a strength for traceability but means we report associations, not fused estimates, and we make no causal claim. RQ-2 is scope-limited to one slice of the trusted list and is preliminary. RQ-3 joins two observatories with divergent coverage, and its decoupling result carries that confound. The assurance-keyword measure in RQ-4 is a lower bound: assurance embedded in a larger contract without an assurance term in the title is missed, though the same is true of the base, so the ratio is conservative in both directions. All figures describe a dated build and will move as coverage deepens.

7. Conclusion

Reading the digital-trust stack from its demand side, we find a state that buys capability ahead of assurance across every layer we can measure. The assured share of procurement is below four percent in both the cryptographic and the artificial-intelligence layers; in the cryptographic layer the assured technology is out-procured by the vulnerable infrastructure it must replace by a factor of about sixty to a hundred, against a fixed migration deadline and a concentrated supply base; and the state is not steering artificial-intelligence procurement toward where harm concentrates.

These are descriptive findings, each traceable to a source the reader can check, and together they make the case that the procurement record is an under-used, provenance-verifiable instrument for observing whether the state is buying the trust it is promising.

Data and code availability

The procurement corpus is on Zenodo, concept DOI 10.5281/zenodo.21192405 (CC-BY-4.0). The cross-observatory analysis, the provenance manifest for the linked datasets, the figure, and this article are at <https://github.com/tyche-dev/trust-tech-procurement-observatory> (**cross-observatory/**). Collection, validation and analysis were performed by the single author; the pipeline is open and deterministic, and any reader can re-harvest and recompute every figure independently. The linked observatories are the qualified-trust-service census and the AI-incident catalogue (Sokolov 2026a); neither is merged into the procurement corpus.

Competing interests

The author declares no competing interests.

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